

Informational Retrieval: Assignment 2 - AI Agent

1. Introduction

In this assignment, I built an artificial intelligence (AI) agent in the form of a chatbot that can (1) remember its past conversations with the user, (2) browse the internet for information, and (3) engage in a philosophical discussion. Although any large language model (LLM) can be used to power the chatbot, my experiment was conducted using the 8 billion parameter version of the Llama Instruct. The model can be run entirely in the terminal or using a Streamlit-powered UI.

2. Development

The models were developed with the help of various artificial intelligence tools, namely ChatGPT, Gemini, Claude, and Codex. The preliminary stage of the development was mainly conducted with the support of ChatGPT. As I did not know how to build an AI agent, I prompted it to guide me through the initial setups. Following that, the agent was further developed with the assistance of Gemini.

In this stage, I mainly brainstormed how to expand the model. That said, the Philosopher Mode was my original idea, meaning the idea itself did not materialize out of the tool's suggestions. It is also in this stage that I added a graphical user interface (GUI).

After I decided that I was satisfied tinkering with the system, I had every piece of code reviewed by Claude. The tools suggested several fixes that made it to the code's final form. These fixes include the accuracy of the safety feature. The codes were then checked for a final review by Codex, who confirmed that I could push it to GitHub.

3. Features

The Philosopher AI Agent is a chatbot that stores its conversations with the user in a compacted memory. Powered by ChromaDB, this feature allows the chatbot to tailor its responses to suit the user. Initially, the system kept encountering errors resulting from too many tokens being sent for processing. Several fixes were introduced to solve this issue, including reducing the amount of memory stored per batch to five memory elements.

The agent is also equipped with the capability to browse the internet using the DuckDuckGo module. It does not browse the internet for every query. Rather, it can intelligently decide whether the query calls for additional information. In cases where the step is unnecessary, it pulls information from the LLM's static memory. In this

case, the LLM in question is the meta-llama/Meta-Llama-3-8B-Instruct, which can always be replaced in the [agent.py](#) code.

A Streamlit-powered UI was added to the system. Other than offering a conventional chatbot experience through the chatbox, this feature allows users to toggle on and off the Philosopher Mode. It also provides a panel through which users can see the memory that the system builds their profile out of.

The main highlight of the agent is the Philosopher Mode. In this mode, any query the user inputs into the system will trigger a discussion on the query. The system will also provide quotes from historical figures. The settings for this mode can be configured in the code [agent.py](#). Currently, the system chooses a random theme for every query out of the following list:

1. Stoicism
2. Islamic philosophy
3. Stoic and grit
4. Quranic wisdom
5. Hadith wisdom
6. Go Rin no Sho
7. Dao de jing
8. Existential and purpose
9. Eastern philosophy
10. Ethics and morality
11. Japanese philosophy
12. Chinese philosophy
13. Psychology